



Innovation & Entrepreneurship Hub for Educated Rural Youth

(SURE Trust - IERY)

TECHNICAL PROJECT REPORT

PROJECT TITLE

RA6M5 Processor Development Board

Domain of the Project

PCB Designing & Embedded Hardware

Team Mentor

Mr. Paramesh

(Mentor – SURE Trust)

Submitted By

Ms. Potnuru Akhila

Project Duration

March 2026 – April 2026

DECLARATION

The project titled “**RA6M5 Processor Development Board**” has been successfully mentored by Mr. Paramesh and organized by **SURE Trust**, with the primary objective of empowering educated rural youth through hands-on industrial project training. This practical exposure bridges academic concepts with real-world employer requirements.

I hereby declare that the work presented in this technical report was carried out by the team member listed below, reflecting genuine effort, high standards of hardware design, and comprehensive technical learning.

Team Member Details:

- **Ms . Potnuru Akhila**

Signature: _____

Mentor's Signature: _____

Name: Mr. Paramesh

Designation: Mentor – SURE Trust
Trust

Executive Director:

Prof. Radhakumari
SURE

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1 Executive Summary

This project centers on the comprehensive design and layout of a high-performance, edge-ready printed circuit board built around the 32-bit Renesas RA6M5 Arm Cortex-M33 microcontroller. The engineering objective was to eliminate messy multi-module breakout prototyping by consolidating system power conditioning, battery charging, state-of-charge tracking, external non-volatile memory, short-range wireless telemetry, and modular plug-and-play sensor interfacing onto a unified, production-ready 4-layer board.

Using KiCad EDA, a full multi-sheet schematic and a compact 94.725\text{ mm} \times 51.7\text{ mm} layout were developed, integrating isolated internal ground and power distribution planes to secure signal integrity and thermal efficiency. The design features USB-C input power, autonomous Li-Po charging via the MCP73831, dynamic battery fuel gauging via the MAX17048, a high-efficiency RT9080-33GJ5 LDO regulator, DA14531MOD Bluetooth Low Energy (BLE) connectivity, high-speed 128 Mb QSPI Flash, power-gated microSD storage, and a SparkFun Qwiic sensor expansion interface. The entire design was verified using strict Electrical Rule Checks (ERC) and Design Rule Checks (DRC), producing error-free manufacturing outputs (Gerber RS-274X, drill files, and BOM) ready for industrial surface-mount fabrication.

2 Introduction

Background and Context of the Project

Modern electronic products in industrial telemetry, smart agriculture, remote health tracking, and home automation require compact embedded nodes that balance low power consumption with high local processing power and wireless communication. While microcontrollers such as the Renesas RA6M5 provide robust processing and peripheral capability, relying on breadboards and scattered expansion shields introduces parasitic inductance, line ringing, signal attenuation, and mechanical connection dropouts. Designing an application-tailored, multi-layer printed circuit board is necessary to translate theoretical circuit designs into reliable industrial hardware

2.1 Problem Statement and Goals of the Project

Contemporary connected hardware typically depends on multiple disparate sub-circuits: battery management, step-down power regulation, wireless radios, external bulk memory, and standard sensor breakout buses. Connecting these blocks with jumper wires increases board surface area, invites ambient electromagnetic interference (EMI), and raises bill-of-materials assembly costs.

The primary goal of this project is to engineer a unified, compact, noise-immune 4-layer embedded development platform housing the Renesas RA6M5 microcontroller alongside all required support subsystems on a single PCB substrate.

2.2 Scope and Limitations of the Project

Scope:

Comprehensive circuit schematic capture divided across modular hierarchical sheets.

Detailed component library selection and footprint association adhering to IPC standards.

Professional 4-layer PCB routing featuring continuous internal reference planes and high-frequency trace management.

Power supply management integration incorporating USB Type-C, single-cell Li-Po charging, battery voltage tracking, and linear regulation.

Design rule validation through automated ERC and DRC, ending in production file generation (Gerbers, Excellon drills, Pick-and-Place, and BOM).

Limitations:

Physical assembly (soldering, surface mounting) and bench bring-up using laboratory instruments

(oscilloscopes, power analyzers) remain outside the scope of this hardware design phase.

Embedded firmware coding, RTOS porting, and software driver deployment are slated for the subsequent phase of the board life cycle.

2.3 Innovation Component in the Project

- **Active Li-Po State-of-Charge Tracking:** Instead of relying on crude analog voltage divider approximations that vary with battery temperature and internal impedance, the design incorporates a dedicated I2C-based fuel gauge (MAX17048) running algorithmic cell tracking.
- **P-MOSFET Controlled Power Gating for MicroSD:** To curb standby parasitic drain during microcontroller deep-sleep intervals, the microSD storage rail is managed through high-side P-channel MOSFET switching, completely turning off peripheral current leakages.
- **Standardized Modular Sensor Expansion:** Integration of a polarized 4-pin SparkFun Qwiic I2C interface permits immediate daisy-chain connection of environmental sensors without soldering or jumper pins.
- **Double-Sided SMT Component Distribution:** By placing larger active semiconductor switches (Q1, Q2) and local decoupling networks on the bottom copper layer, the top surface maintains clean breakout clearances for high-density header connectors, user switches, and status indicators.

3 Project Objectives and Methodology

3.1 Clearly Defined Objectives and Goals of Project

- Design an industrial-grade embedded computing board around the 32-bit Renesas RA6M5 microcontroller using KiCad EDA.
- Create a hierarchical circuit schematic covering power delivery, processing, storage, RF wireless communication, and debug interfaces.
- Route a 4-layer printed circuit board observing high-speed mixed-signal layout rules, continuous return planes, and short bypass loops.
- Integrate a complete power infrastructure comprising USB Type-C, an MCP73831 lithium charger, an RT9080 low-dropout regulator, and MAX17048 fuel monitoring.
- Provide external storage architectures through a 128 Mb QSPI flash memory chip and a dedicated microSD card slot.
- Establish wireless short-range communication using the DA14531MOD Bluetooth Low Energy transceiver module with a clear keepout boundary.
- Validate design integrity by resolving all electrical and physical rule clearance conflicts (zero ERC and DRC errors) and generating manufacturing Gerbers.

3.2 Expected Outcomes and Deliverables

- Complete, verified KiCad schematic package (.kicad_sch) with structured sub-sheets.
- Fully routed and DRC-verified 4-layer PCB layout file (.kicad_pcb).
- Production fabrication files containing industry-standard Gerber RS-274X layers and NC drill files.
- Complete Bill of Materials (BOM) detailing reference designators, component values, part packages, and manufacturer ordering codes.
- 3D spatial models (.step) confirming clear mechanical tolerances and correct component footprints.

3.3 Methods and Tools Used

- **Methods/Technology used:** The project was carried out using a systematic ap-proach to design a professional embedded development board, ensuring the final PCB design was reliable, manufacturable, and suitable for IoT and embedded sys-tem applications. The design process followed a structured multi-step methodology, including requirement analysis, component selection,

schematic design with electrical checks, 4-layer PCB layout routing, design rule verification, and manufacturing output generation.

- **Tools/Software used:** The project was designed using KiCad, an open-source Electronic Design Automation (EDA) software. KiCad served as the primary environment for schematic design, PCB layout, design verification (ERC and DRC), 3D visualization, and generating Gerber and BOM files required for PCB manufacturing.
- **Data collection approach:** Component parameters, electrical specifications, and functional layout requirements were gathered from industry-standard datasheets and application notes to ensure proper compatibility, performance, and hardware reliability.

4 Project Architecture and Working Screenshots Ex-planation

4.1 Project Architecture & Flow Diagrams

The system follows a structured modular architecture. Below are three different visual representations showcasing the power distribution, core microcontroller processing, and peripheral subsystem integration.

4.1.1 Flowchart Style 1: Graphical Block Architecture (TikZ)

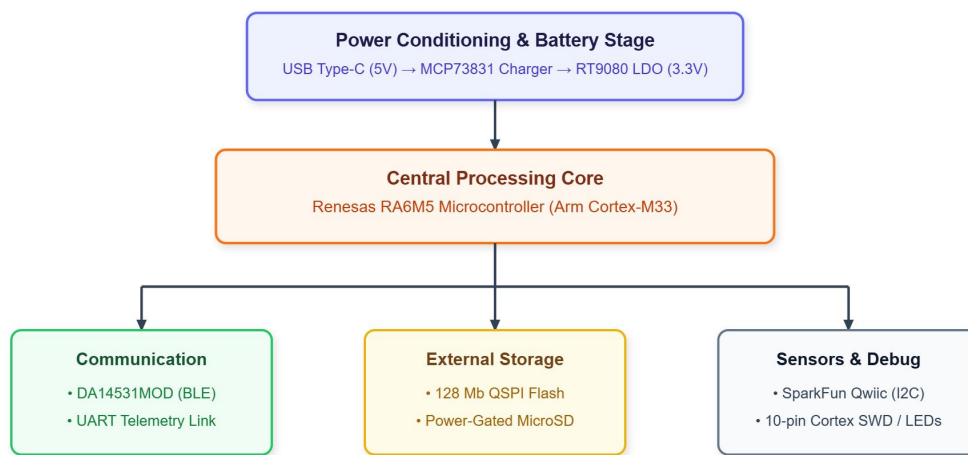


Figure 1: System Architecture Flowchart - Style 1 (Graphical Blocks)

4.1.2 Flowchart Style 2: Structured Box Architecture (TikZ Layout)

4.1.3 Flowchart Style 3: Sequential Linear Pipeline Architecture

- **Power Flow & Regulation:** External power is supplied via the USB Type-C connector, which feeds the battery charging circuitry and regulators to provide steady voltage rails across the board.
- **Core Processing Unit:** The Renesas RA6M5 microcontroller serves as the central intelligence, coordinating all internal data operations and peripheral interfaces.
- **Peripheral Subsystem Distribution:** The architecture groups external components logically into communication links, memory storage arrays, and expansion or debugging interfaces to ensure clean and organized hardware integration.

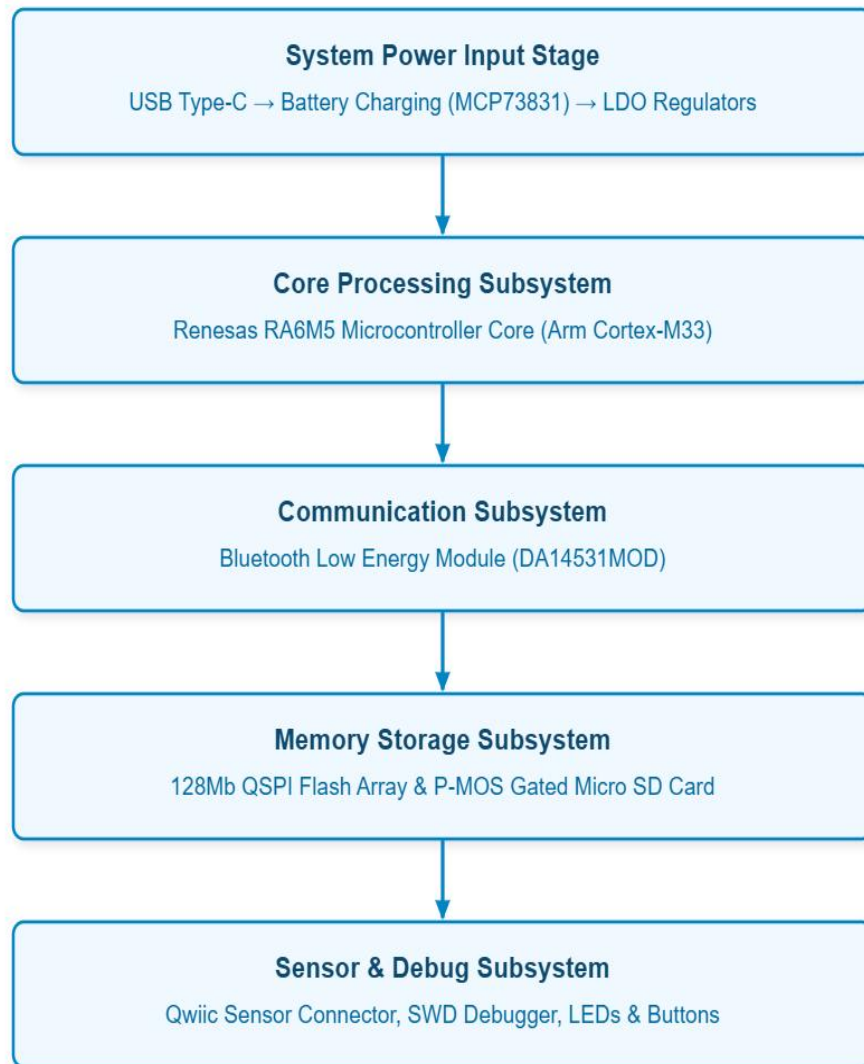


Figure 2: System Architecture Flowchart - Style 2 (Structured Box Layout)

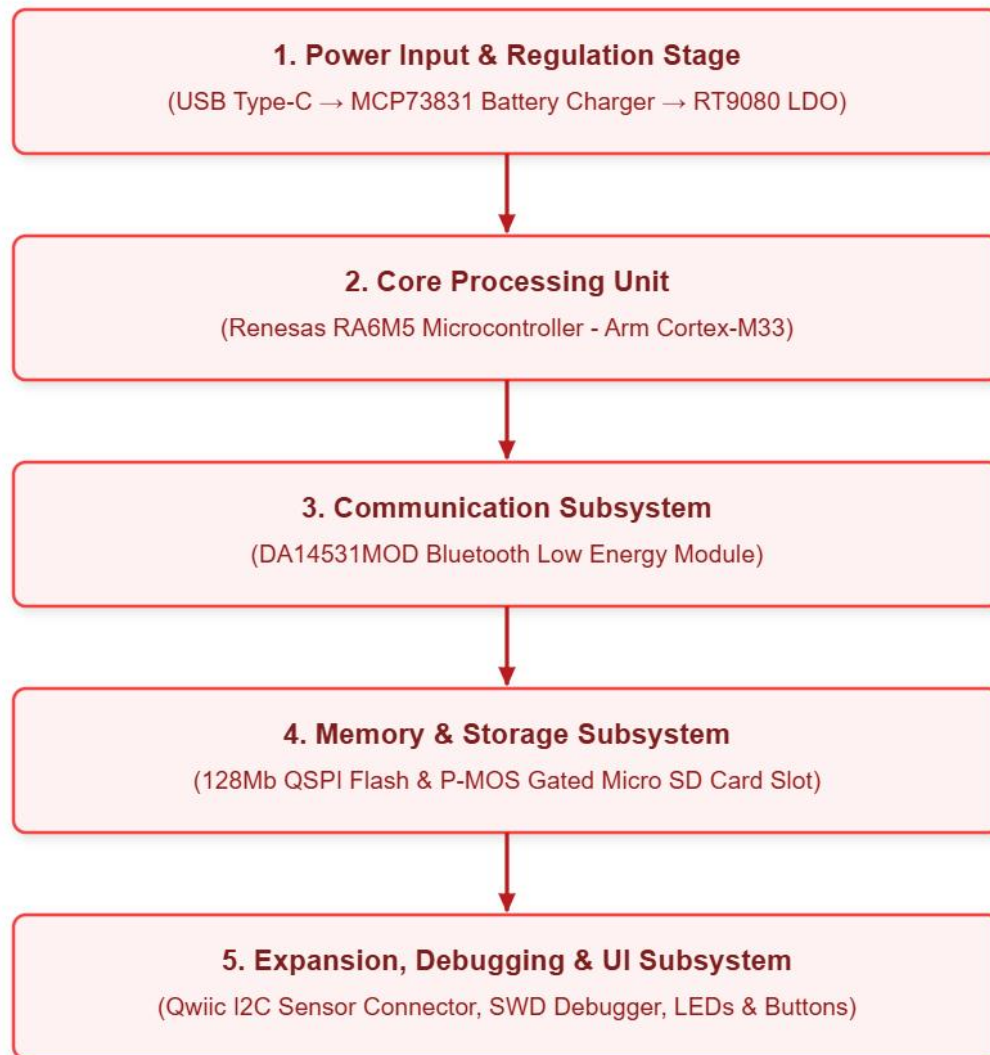


Figure 3: System Architecture Flowchart - Style 3 (Sequential Pipeline)

4.2 Final Project Working Screenshots & Visual Verification

4.2.1 Complete Schematic Diagram

The hierarchical schematic captures all functional blocks—ranging from the power regulation sheet to the main controller sheet—verified thoroughly through Electrical Rule Checks (ERC) to ensure zero open nodes or connection conflicts.

- The power supply and regulation sheet details the USB-C input stage, ESD protection, LM66200/RT9080 regulators, MCP73831 battery charger, and MAX17048 fuel gauge circuits.
- The main controller sheet captures the Renesas RA6M5 microcontroller core, digital GPIO routing, analog input conditioning, crystal oscillators, and communication headers.

- Root hierarchical sheet interconnections establish the top-level link between the modular power management sub-sheet and main processor sheet via standardized bus lines.

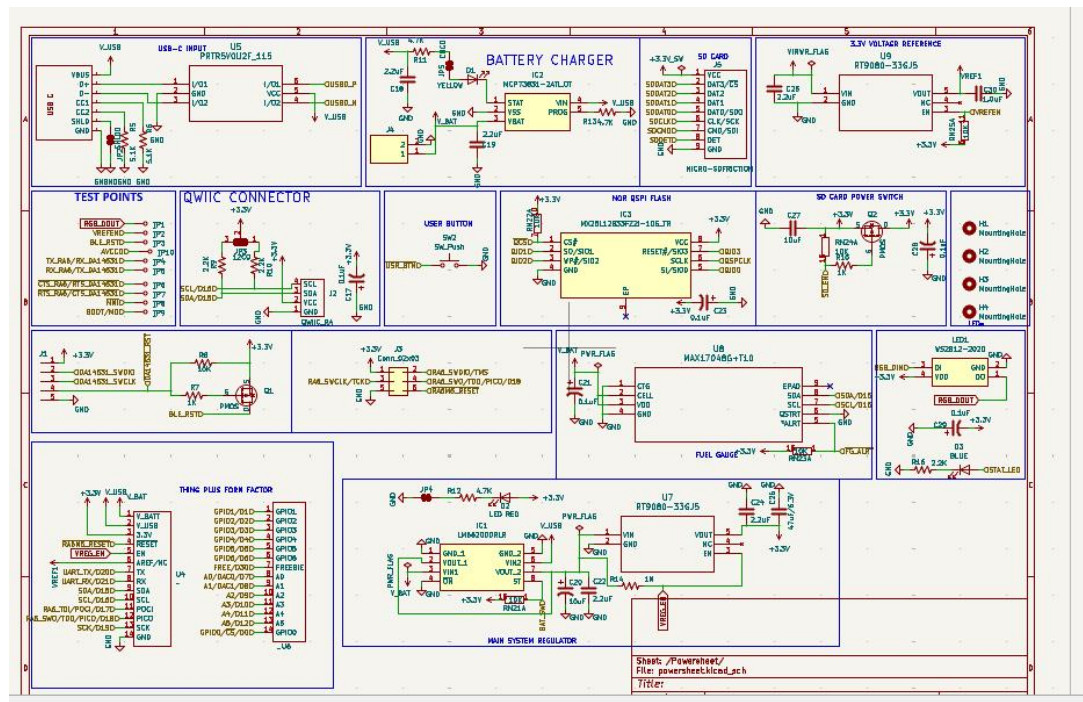


Figure 4: Power Management and Subsystem Schematic Sheet

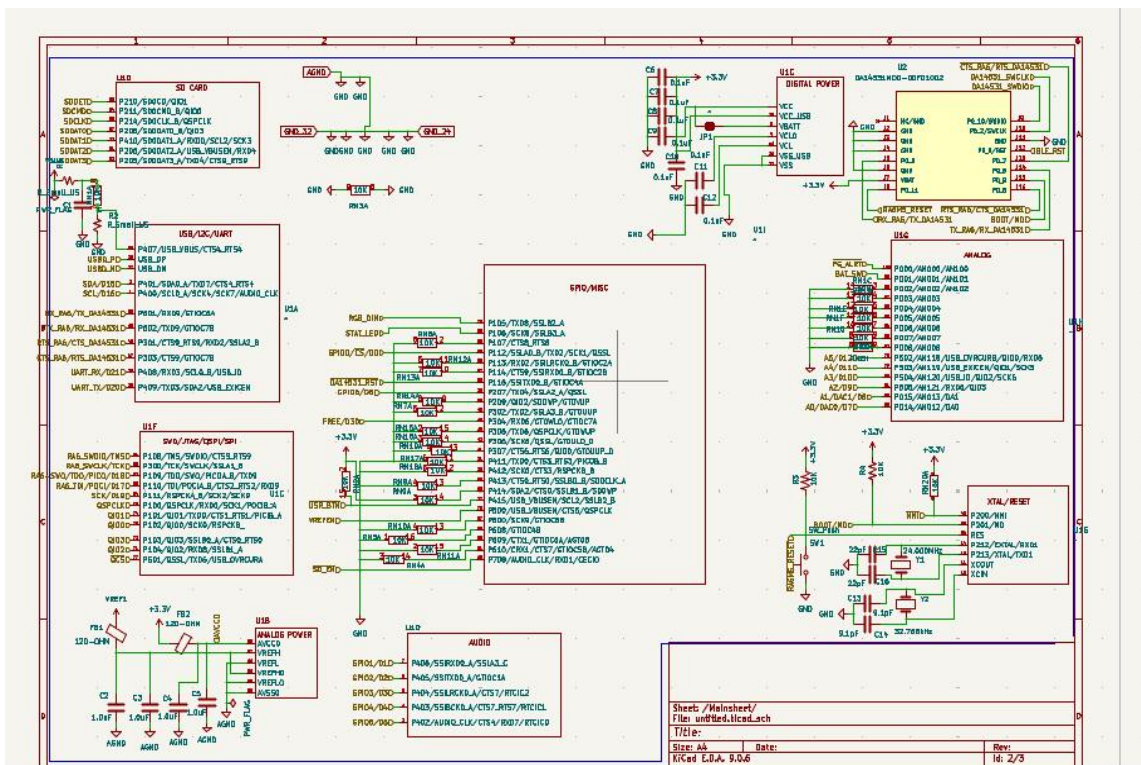


Figure 5: Main Microcontroller and Peripherals Schematic Sheet

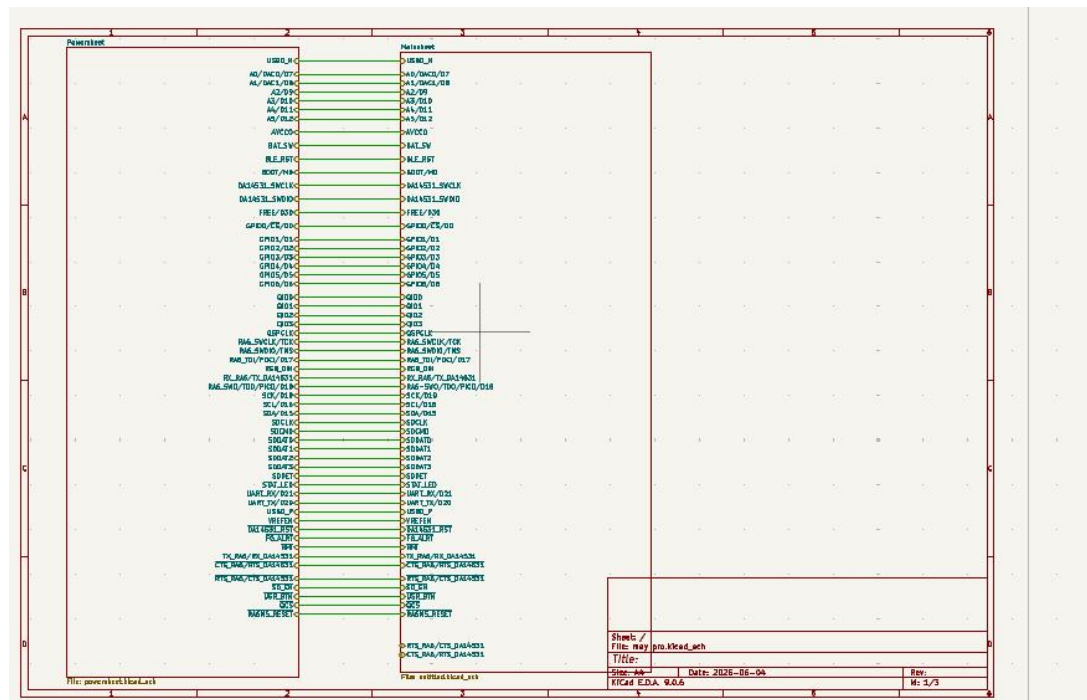


Figure 6: Root Hierarchical Interconnections Schematic Sheet

4.2.2 PCB Layout

The multi-layer board view illustrates the precise 4-layer stackup implementation, showing how component placement, high-speed signal routing on external copper layers,

internal power planes, and dedicated ground pours were executed in KiCad and validated via Design Rule Checks (DRC).

- The combined multi-layer routing layout view displays top-layer traces, internal power planes, bottom copper ground pours, and design rule compliance across the board.
- Dedicated front assembly and top trace configurations optimize high-speed signal paths while maintaining safe component clearances.
- Bottom-layer trace routing paths, vias, and continuous ground plane coverage provide effective noise shielding and stable power return paths.

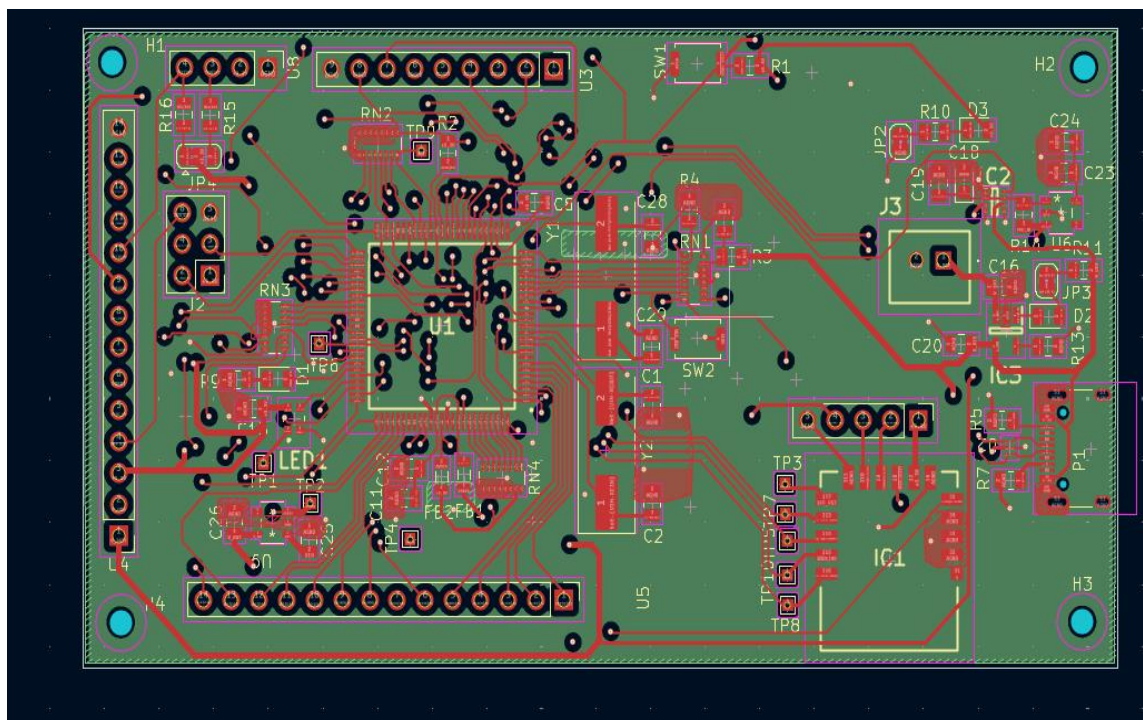


Figure 7: PCB Layout - Top Copper Layer Routing and Placement

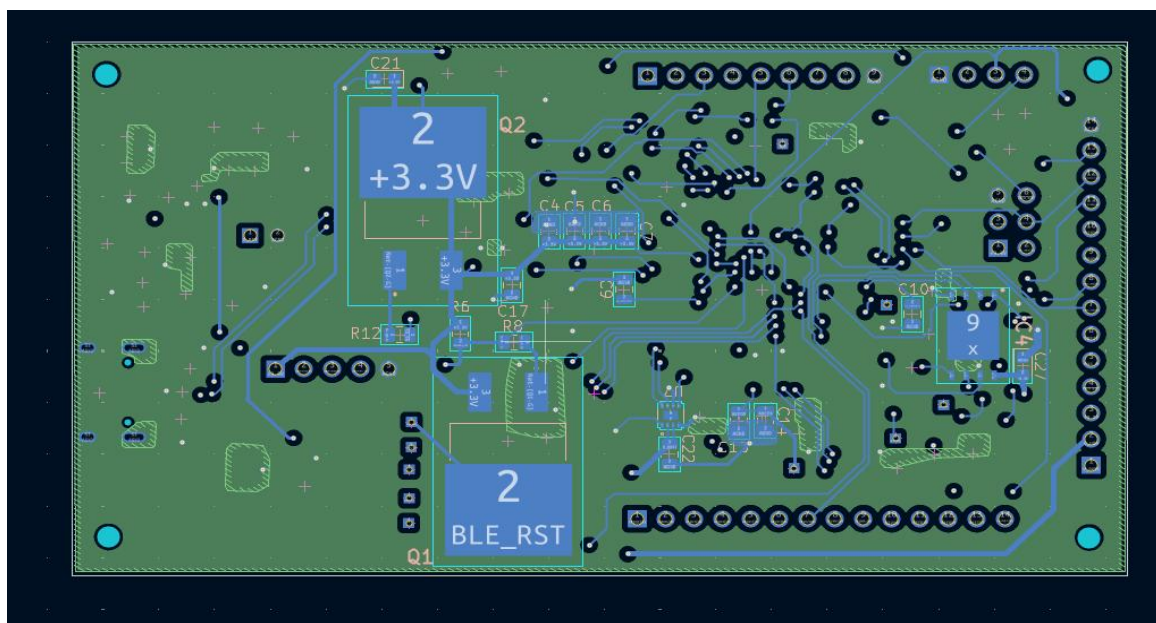


Figure 8: PCB Layout - Bottom Copper Layer Routing and Traces

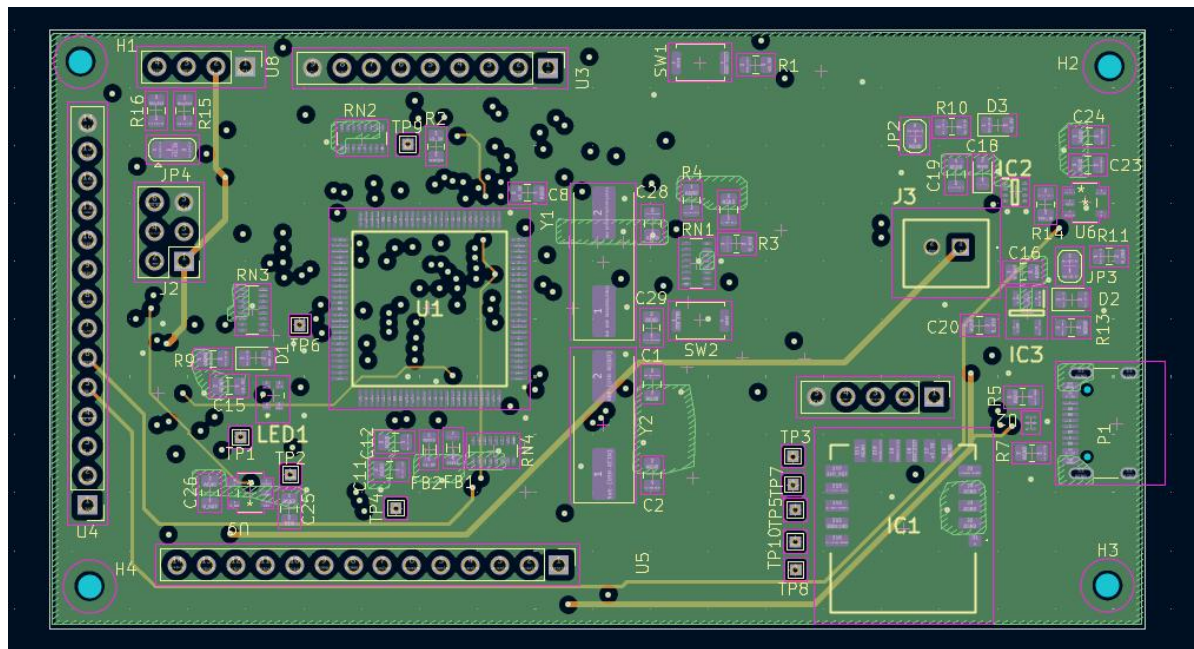


Figure 10: PCB Layout - Internal Power Plane View

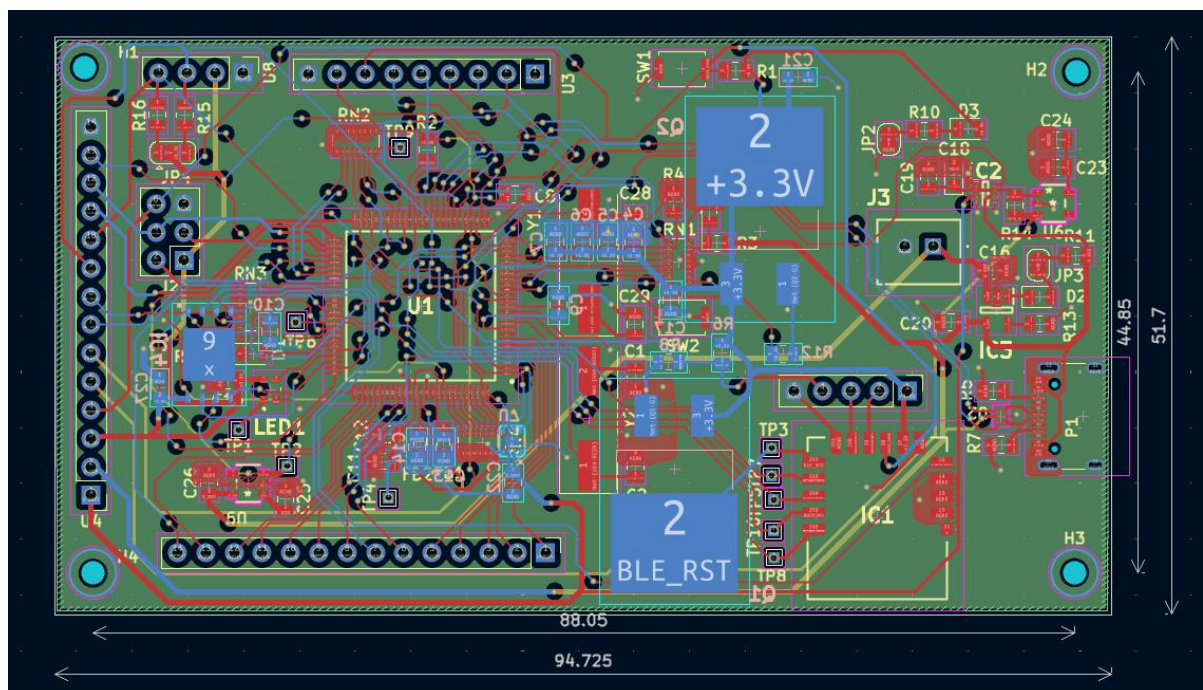


Figure 11: PCB Layout - Combined Multi-Layer Routing View

4.2.3 3D View of the PCB

The realistic 3D rendered view provides physical verification of the assembled board, confirming proper component heights, correct connector orientations, silkscreen readability, and mechanical clearance for the mounting holes.

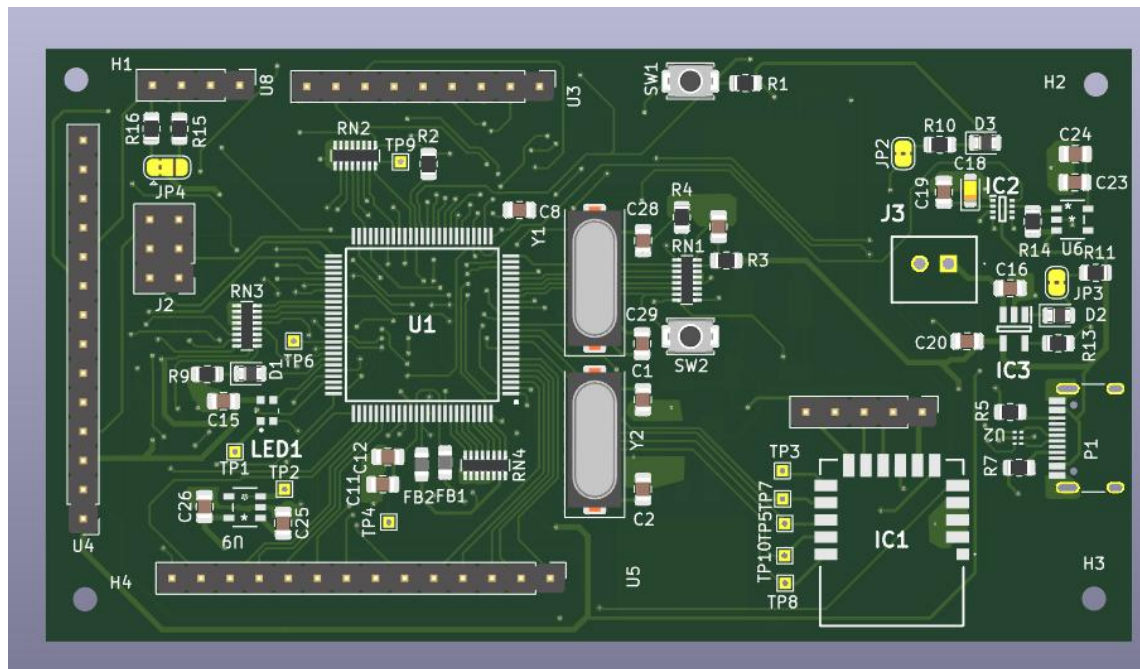


Figure 12: 3D Rendered View of the PCB - Top Assembly and Component Placement

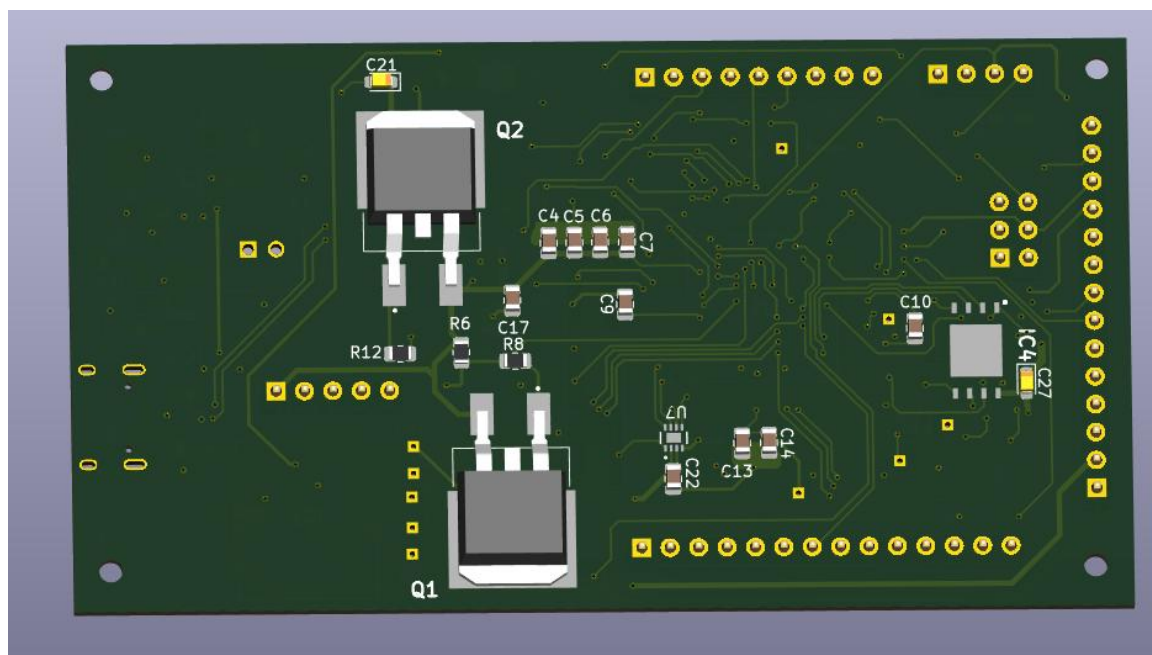


Figure 13: 3D Rendered View of the PCB - Bottom View and Trace Routing Verification

5 Social / Industry Relevance of the Project

- **Industrial Hardware Modernization:** By integrating battery management, high-density storage, wireless telemetry, and sensor connectivity onto a single PCB, this board eliminates bulky multi-shield assemblies, significantly lowering bill-of-materials and manufacturing costs for production IoT systems.
- **Edge Computing and Predictive Maintenance:** The combination of an Arm Cortex-M33 core and high-speed QSPI flash allows vibration, temperature, and current data to be sampled, stored, and analyzed directly on edge nodes in industrial environments.
- **Remote Healthcare and Environmental Monitoring:** The low-power BLE module and dedicated fuel gauge facilitate long-term standalone operation in field locations, making the hardware well suited for continuous environmental and biometric sensing applications.
- **Bridging Rural Talent with Electronics Manufacturing:** Developed under SURE Trust's rural empowerment initiative, this project equips emerging engineers with direct experience in industry-standard design workflows, PCB stackup planning, and design-for-manufacturing (DFM) principles.

6 Learning and Reflection

6.1 Document Every Team Member's New Learnings

- **Multi-Layer PCB Layout:** Acquired hands-on proficiency in structuring and routing a 4-layer PCB, isolating signal traces, and utilizing dedicated inner solid copper ground and power planes.
- **Power Management Architecture:** Mastered the circuit design of autonomous single-cell Li-Po battery charging stages, low-dropout linear voltage regulators, and I2C fuel-gauge measurement subsystems.
- **High-Speed Memory & Bus Routing:** Learned design rules for routing high-speed QSPI flash lines, SPI interfaces, and SWD debug links while matching trace lengths and preserving continuous reference return planes.
- **RF Integration and Keepouts:** Learned keepout layout requirements for RF antennas, ensuring no ground pours or copper traces interfere with the DA14531MOD wireless module.
- **Design Rule Verification & DFM:** Gained thorough experience running Electrical Rule Checks (ERC) and Design Rule Checks (DRC), resolving clearance errors, and exporting fabrication-ready Gerber RS-274X datasets.

6.2 Document Every Team Member's Experience

Working on this project from initial block diagrams to final manufacturing files provided invaluable practical experience in hardware engineering. Managing high routing density around the 100-pin RA6M5 microcontroller and routing the power traces on a 4-layer board required careful component placement and systematic via planning. Overcoming design rule check conflicts and addressing copper clearance constraints deepened my understanding of industry design-for-manufacturing standards. This journey with SURE Trust bridged the gap between academic electronics theory and production-grade embedded hardware design.

7 Conclusion and Future Scope

7.1 Recap Objectives and Achievements

All technical objectives defined at the start of the project were successfully achieved. A unified, 4-layer embedded development board centered on the Renesas RA6M5 Arm Cortex-M33 microcontroller was designed, routed, and verified using KiCad EDA. The platform combines a USB-C input stage, autonomous battery charging, fuel-gauge monitoring, BLE wireless communication, 128 Mb QSPI flash, a power-gated microSD card interface, and an external SparkFun Qwiic connector onto a single board measuring $94.725\text{ mm} \times 51.7\text{ mm}$. The design passed all ERC and DRC routines without violations, yielding complete Gerber, drill, and BOM files ready for fabrication.

7.2 Future Scope of this Project

- **Physical Prototyping & Bring-Up:** Fabricating the physical PCB prototypes, carrying out surface-mount assembly, and inspecting voltage rail stability, crystal oscillation, and reset responses using digital storage oscilloscopes.
- **Bare-Metal Firmware & Driver Development:** Writing modular C/C++ board support packages (BSP) for the Renesas RA6M5 to initialize QSPI flash read/write cycles, handle microSD FAT32 operations, and run BLE telemetry advertising.
- **Dynamic Low-Power Tuning:** Measuring active, idle, and deep-sleep currents under varying processor load conditions to optimize battery operating lifetimes in field deployments.
- **Plug-and-Play Sensor Expansion:** Connecting multiple environmental sensors via the onboard SparkFun Qwiic interface to establish localized industrial telemetry nodes.
- **Protective Enclosure Design:** Modeling and 3D-printing a rugged, snap-fit enclosure tailored to the $94.725\text{ mm} \times 51.7\text{ mm}$ board outline, providing access to top-side pushbuttons and edge connectors.